CoProU-VO: Combining Projected Uncertainty for End-to-End Unsupervised Monocular Visual Odometry

Jingchao Xie * [1 3] Oussema Dhaouadi * [1 2 3] Johannes Meier [1 2 3] Jacques Kaiser [2] Daniel Cremers [1 3]

[1] TUM [2] DeepScenario [3] MCML Munich Center for Machine Learning

Overview

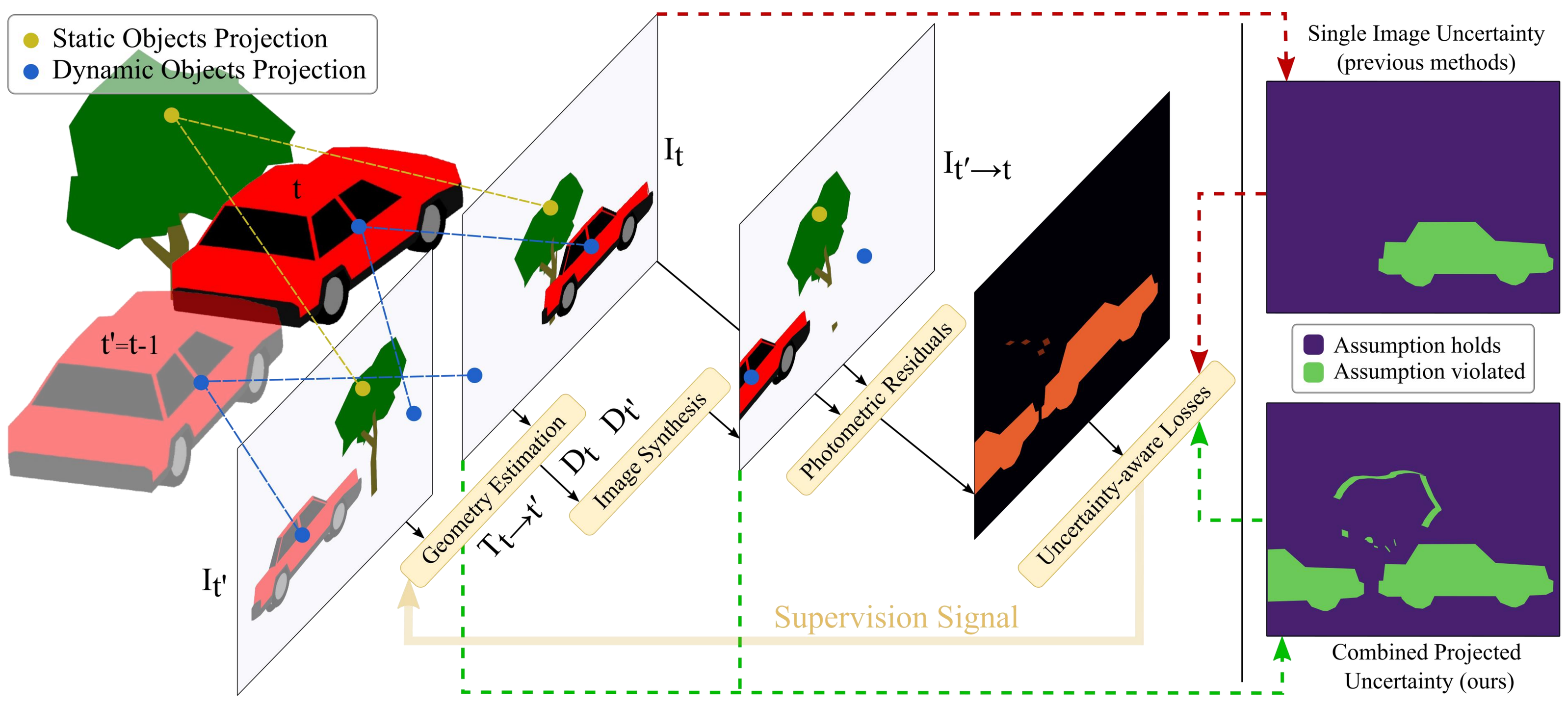
Problem Statement

Unsupervised visual odometry (VO) relies on static scene assumption. However, this assumption is violated by reflective surfaces, occlusions, and dynamic objects.
➔ Uncertainty modeling can help.

Reflective surfaces *Occlusions* *Dynamic objects*

Contribution

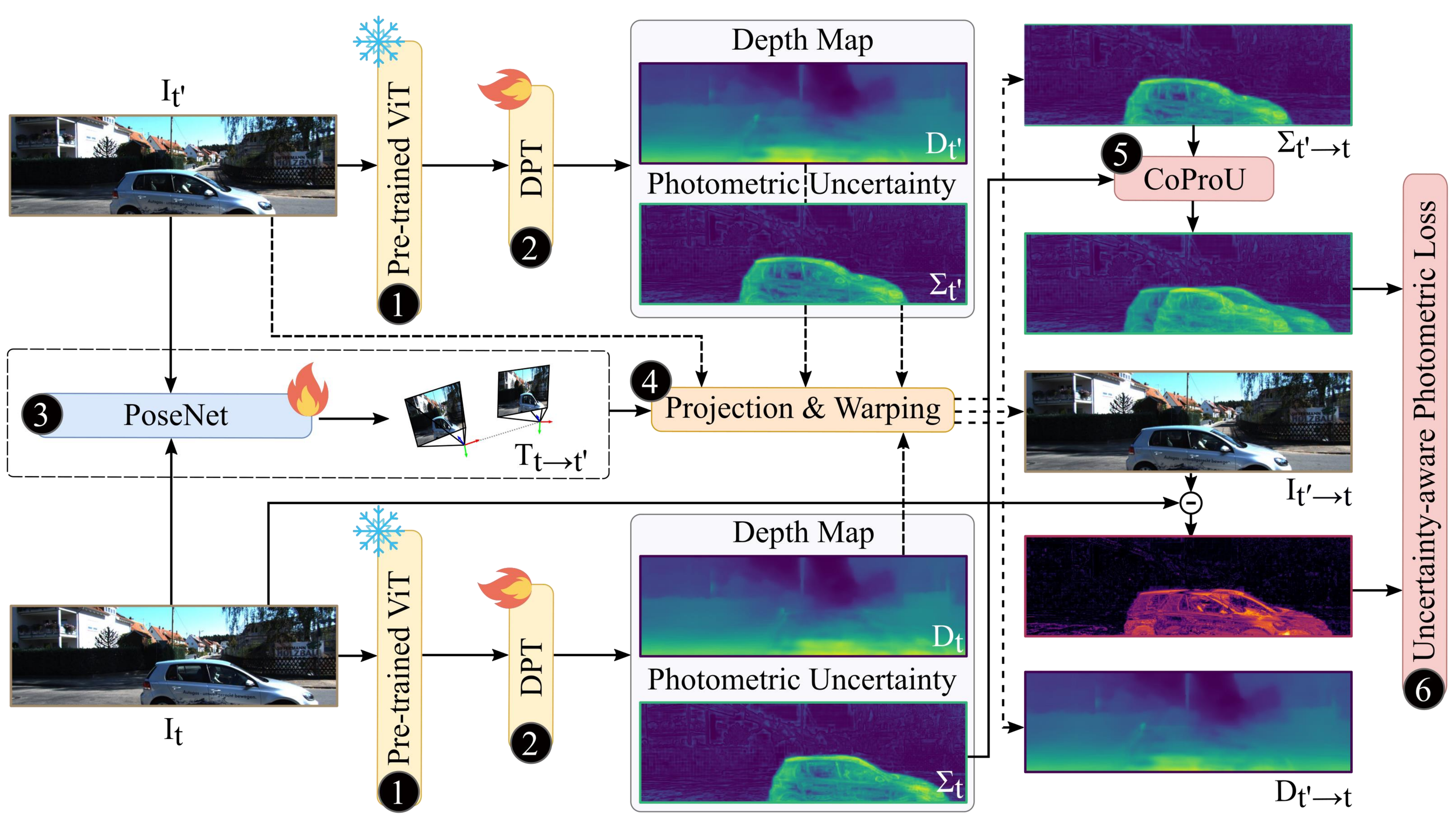
How do prior methods handle uncertainty?
They model only target-frame uncertainty, ignoring errors from reference frames.
How do we handle uncertainty?
CoProU-VO: combines target and projected reference uncertainty.



***Contribution:** Prior methods mask only target-image uncertainty (red). Our approach combines target and reference uncertainties (green) for robust masking of dynamic regions and occlusions.*

Propagating uncertainty, improving odometry

Method



***CoProU-VO Overview:** (1) Feature extraction, (2) depth & uncertainty prediction, (3) pose estimation, (4) view synthesis, (5) CoProU combines target-reference uncertainties, (6) uncertainty-aware loss.*

Our CoProU-VO framework takes two consecutive frames as input:

1. **Feature Extraction:** A pretrained vision transformer backbone extracts features.
2. **Depth & Uncertainty :** A decoder predicts depth and uncertainty for both frames.
3. **Pose Estimation:** A PoseNet module estimates the relative camera pose.
4. **View Synthesis:** Projection and warping operations reconstruct target views from reference frames.
5. **Uncertainty Integration:** The CoProU module combines uncertainties from target and reference frames.
6. **Loss Computation:** An uncertainty-aware loss supervises training for robust VO in dynamic scenes.

Unlike prior methods that only consider target-frame uncertainty, CoProU integrates cross-frame uncertainties, enabling robust handling of occlusions and dynamic objects

Project & Combine Uncertainty

1. Predict uncertainty for both frames.
2. Project reference uncertainty onto the target view using the same warping as image synthesis:

$$\Sigma_{t'\to t}(p_t) = warp(\Sigma_{t'}, T_{t\to t'}, D_t, K)$$

3. Combine target and projected reference uncertainties into a single term:

$$\sigma_{eff}(p_t) = \sqrt{\Sigma_t(p_t)^2 + \Sigma_{t'\to t}(p_t)^2}$$

4. This effective uncertainty downweights unreliable regions (dynamic objects, occlusions, reflective surfaces) in the photometric loss:

$$\mathcal{L}_P = \frac{1}{|V|}\sum_{p\in V}\frac{r\left(I_t(p), I_{t'\to t}(p)\right)}{\sigma_{eff}(p)} + \log \sigma_{eff}(p)$$

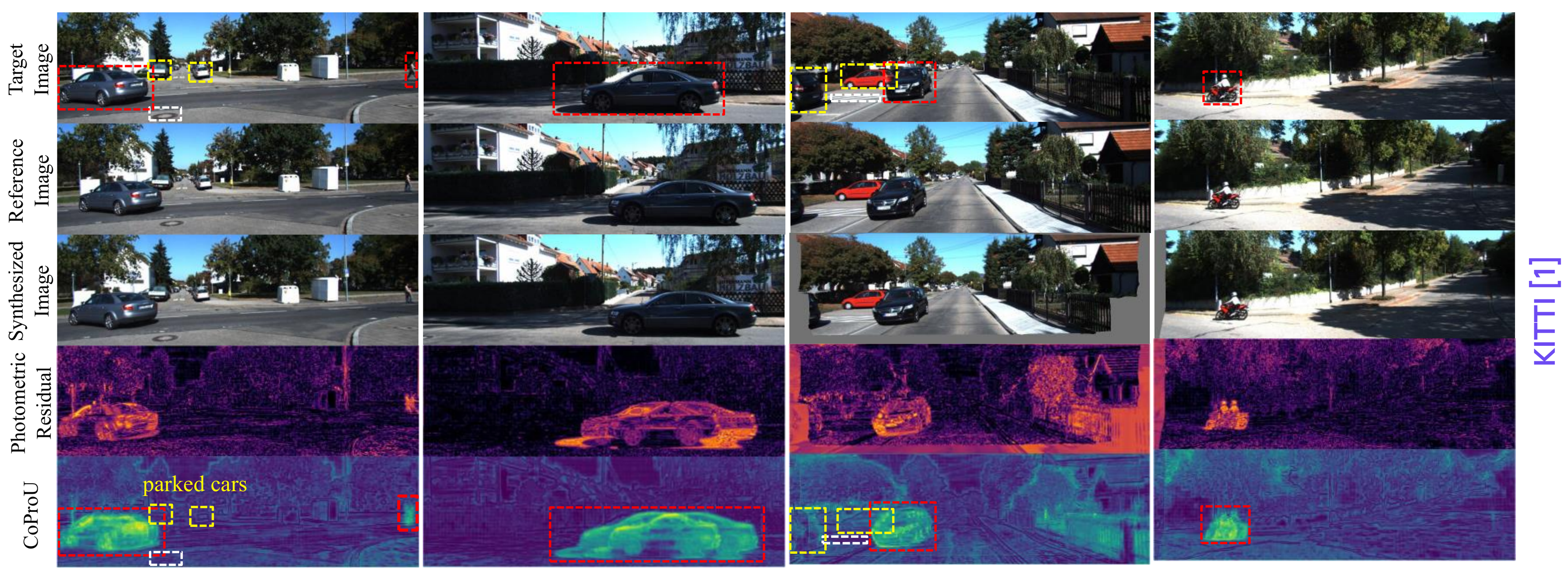
Results

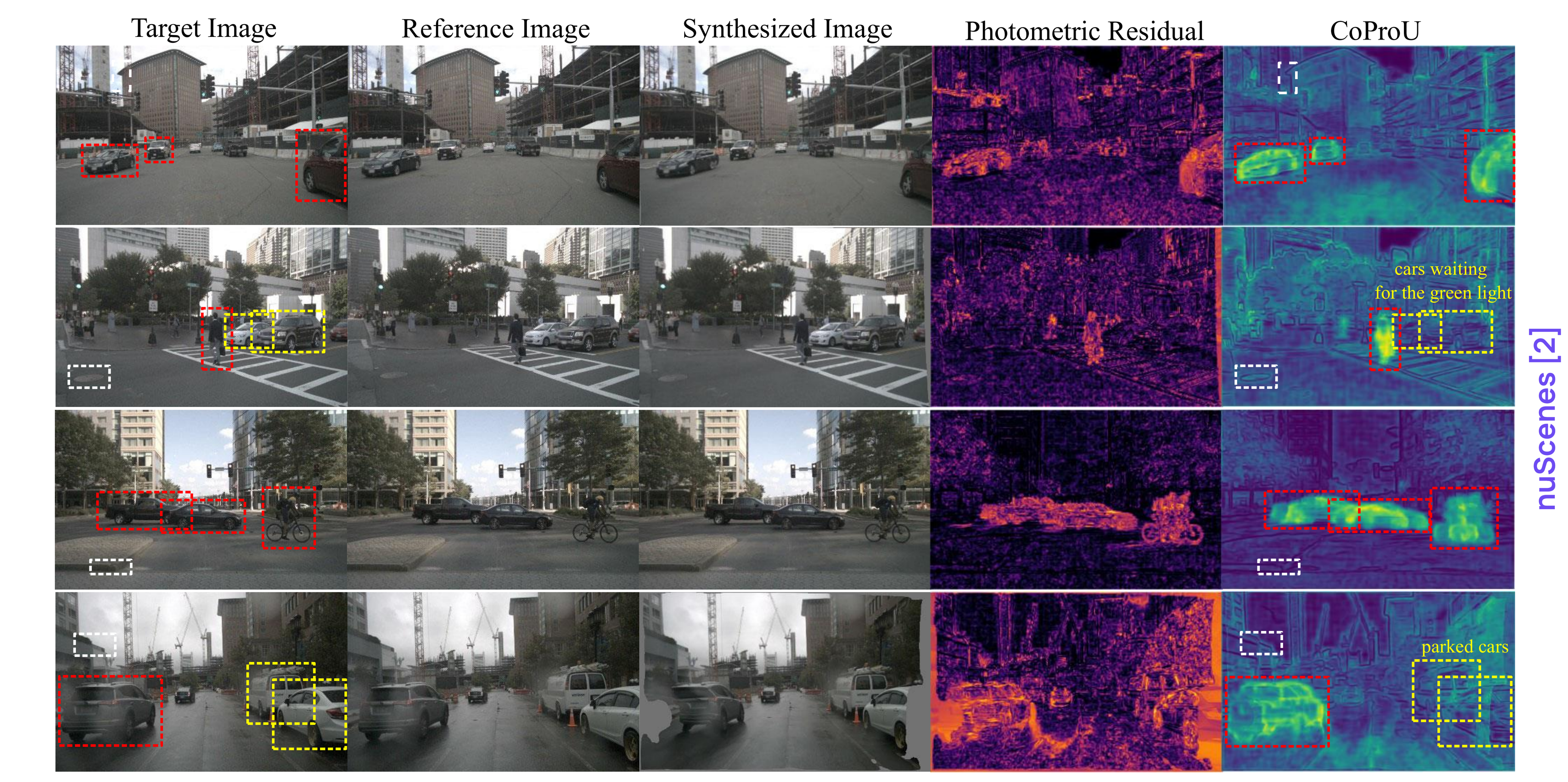
Quantitative Results

Methods	Seq. 01			Seq. 09			Seq. 10		
	ATE	t_{err}	r_{err}	ATE	t_{err}	r_{err}	ATE	t_{err}	r_{err}
SfMLearner [3]	109.61	22.41	2.79	77.79	19.15	6.82	67.34	40.40	17.69
Depth-VO-Feat† [4]	203.44	23.78	1.75	52.12	11.89	3.60	24.70	12.82	3.41
MonoDepth2 [5]	-	-	-	76.22	17.17	3.85	20.35	11.68	5.31
Zou et al.‡ [6]	-	-	-	<u>11.30</u>	**3.49**	**1.00**	<u>11.80</u>	**5.81**	**1.80**
SC-Depth (Baseline) [7]	313.86	87.04	**1.17**	26.86	7.80	3.13	13.00	7.70	4.90
Manydepth2 [8]	-	-	-	-	7.01	<u>1.76</u>	-	<u>7.29</u>	<u>2.65</u>
CoProU-VO + DINOv2	**63.73**	**19.61**	<u>1.54</u>	14.01	4.70	1.89	13.46	7.64	3.67
CoProU-VO + DepthAnythingV2	<u>75.27</u>	<u>22.08</u>	1.99	**9.84**	<u>4.56</u>	2.02	**11.28**	7.76	3.58

***End-to-end visual odometry results on KITTI [1]:** Monocular two-frame training unless † (stereo) or ‡ (video). Best in **bold**, second-best <u>underlined</u>. Metrics: ATE [m], t_{err} [%], r_{err} [°/100m].*

Qualitative Results





***Uncertainty Visualization:** Our CoProU detects dynamic objects (red) with high uncertainty, marks static objects (yellow) as certain, and assigns low uncertainty to informative features (white), highlighting reliable regions for pose estimation.*

Conclusion

CoProU-VO **combines uncertainties from both target and reference frames** to robustly mask **dynamic and occluded regions**, achieving **improved visual odometry** performance in challenging dynamic environments.

References
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